Integration Specification

API Specification for the Easy POS – Buzzebees Integration

Membership, loyalty points and coupons at the point of sale in every Oshinei branch

Party	Role in the project	Main scope of work
Buzzebees	Developer and system operator	Builds APIs B0–B9, decodes the receipt QR, provides the sandbox, LINE Official Account and the reconciliation report
Easy POS	Specification owner and POS configurator	HO_CRM connector specification, receipt QR format, per-branch URL and API key setup, coupon-to-discount-code mapping
Oshinei	Project owner, business-rule owner and acceptance authority	Earn and coupon rules, pilot branches, cashier training, daily reconciliation and acceptance

> **Document status: Draft 1.1** — All field names, endpoints and timeouts are proposals. Buzzebees will align them with the Easy HO_CRM specification once received. Values marked "(proposed)" must be confirmed by the responsible party before use.

Prepared by Oshinei · 14 October 2026 · This document is the working paper for the joint meeting of Oshinei, Easy POS and Buzzebees. The Thai edition is the reference version.

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How to read: sections 1–5 for all parties · sections 6–10 for the Buzzebees and Easy technical teams · sections 11–12 for the acceptance and decision meetings.

1. Definitions

The following terms carry the same meaning throughout this document.

Term	Meaning
API (Application Programming Interface)	An interface through which one system calls the services of another over the network
Endpoint	The URL and HTTP method of a single service, e.g. POST /pos/v1/points/earn
Earn	The customer receives points for the purchase amount of a bill
Burn (redemption)	The customer uses points as a discount on a bill
Coupon	A single-use discount entitlement with a unique code, issued by Buzzebees
Bill key	branch_code + sale_date + receipt_no, identifying one bill, because receipt numbers repeat across branches and days
request_id	A unique request reference generated by the POS, used to retry a request without creating a duplicate transaction
transaction_ref	A transaction reference generated by Buzzebees, used for voids (B6), status queries (B8) and reconciliation (B9)
Idempotency	The property that repeating the same request any number of times yields the same result, with no duplicate deduction of points or coupons
Timeout	The maximum time the POS waits for a response; beyond it the request is treated as failed
HO_CRM	The external CRM connector built into Easy POS, through which the POS calls the Buzzebees APIs
Receipt QR	A QR code printed on every receipt so the customer can claim points through the LINE Official Account
Void	Cancellation of an entire bill after it has been closed
Reconciliation	Comparison of the Buzzebees and Easy records to confirm that both sides agree
Sandbox	A test environment separate from production, used for development and testing

2. Scope and roles

The system shall allow cashiers to look up members, record point earning, apply coupons, burn points and void transactions at the point of sale of every branch running Easy POS (approximately 35 branches). Member sign-up takes place only in the Buzzebees LINE Official Account; the POS does not support sign-up.

Party	Role	Responsibilities
Buzzebees	Developer and system operator	Develop APIs B0–B9 to the HO_CRM specification; decode the receipt QR; provide the sandbox and test accounts; deliver the daily reconciliation report; configure the LINE Official Account, earn rules and coupons
Easy POS	Specification owner and POS configurator	Deliver the HO_CRM specification and receipt QR format to Buzzebees; configure per-branch URLs and API keys in the POS; enable the receipt QR per branch; map coupons to Easy discount codes; take part in testing
Oshinei	Project owner, decision maker and acceptance authority	Set earn and coupon rules; select pilot branches; train cashiers; perform daily reconciliation; accept the work

Out of scope: replacing the POS with B-POS; online ordering and delivery; member sign-up at the POS; marketing campaigns (handled inside Buzzebees with no API impact).

3. Responsibility for each task

Buzzebees develops and operates the system, Easy provides the specification and configures the POS, and Oshinei sets the business rules and accepts the work. Any task without an owner in this table is unassigned and must be assigned at the joint meeting.

No.	Task	Responsible	Supporting / reviewer	Timing
1	Deliver the HO_CRM specification and receipt QR format	Easy (technical team)	Buzzebees receives	Before development
2	Align field names and endpoints in this document with the specification	Buzzebees (development lead)	Easy and Oshinei review	Before development
3	Develop APIs B0–B9 and provide the sandbox	Buzzebees (development team)	Oshinei accepts	Development
4	Configure per-branch URLs and API keys; enable HO_CRM and the receipt QR in the POS	Easy (technical team)	Buzzebees issues API keys	Before testing
5	Map each coupon type to an Easy discount code	Easy (configuration)	Oshinei defines the list	Before testing
6	Define earn and coupon rules and configure them in the back office	Oshinei (project owner)	Buzzebees configures	Before testing
7	LINE Official Account sign-up page and PDPA consent wording	Buzzebees	Oshinei approves wording	Before testing
8	Sandbox testing against the test cases (section 11)	Buzzebees with Easy	Oshinei accepts	Testing
9	Pilot-branch testing with live bills	Easy (field team) with Buzzebees	Oshinei (branch operations)	Pilot
10	Cashier training and screen guide	Oshinei (branch operations)	Easy provides screenshots	Before go-live
11	Daily reconciliation and reporting of mismatches	Oshinei (accounting)	Buzzebees corrects within 3 business days	After go-live, daily
12	Operations: incidents, key rotation, API versioning	Buzzebees (operations)	Easy and Oshinei are notified	After go-live
13	Data Processing Agreement (DPA) and data policy	Oshinei (data controller)	Buzzebees (data processor)	Before go-live

Key contacts (names to be completed)

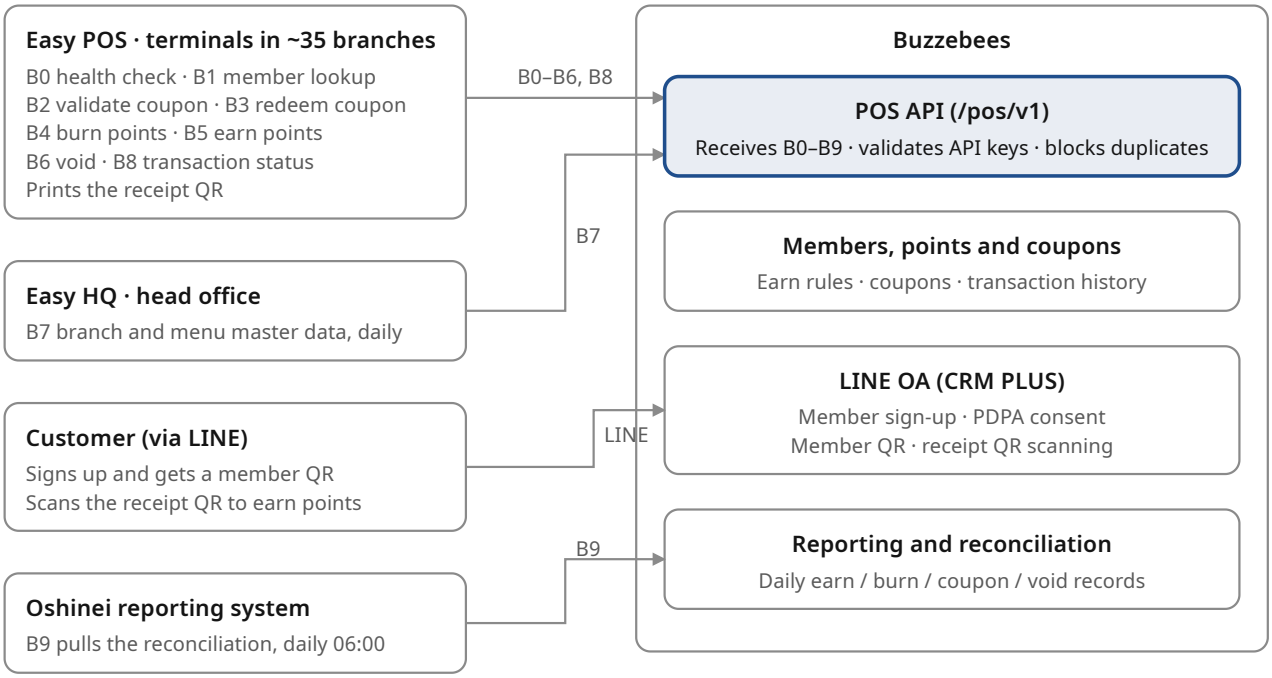
Party	Position	Name	Contact
Buzzebees	Project manager		
Buzzebees	API development lead		
Easy POS	Account manager		
Easy POS	HO_CRM technical team		
Oshinei	Project owner		
Oshinei	Accounting (reconciliation)		
Oshinei	Branch operations (pilot and training)		

4. API overview and integration map

Provider-side developer: Buzzebees · Callers: Easy POS (B0–B8), Easy HQ (B7), Oshinei reporting system (B9)

Every API is called by a consumer towards Buzzebees. Buzzebees never calls the POS, because POS terminals sit inside the branch network and cannot accept inbound connections.

Every call goes to Buzzebees: Easy POS 8 APIs, Easy HQ 1, Oshinei 1



Integration map: four callers and the Buzzebees components that serve them

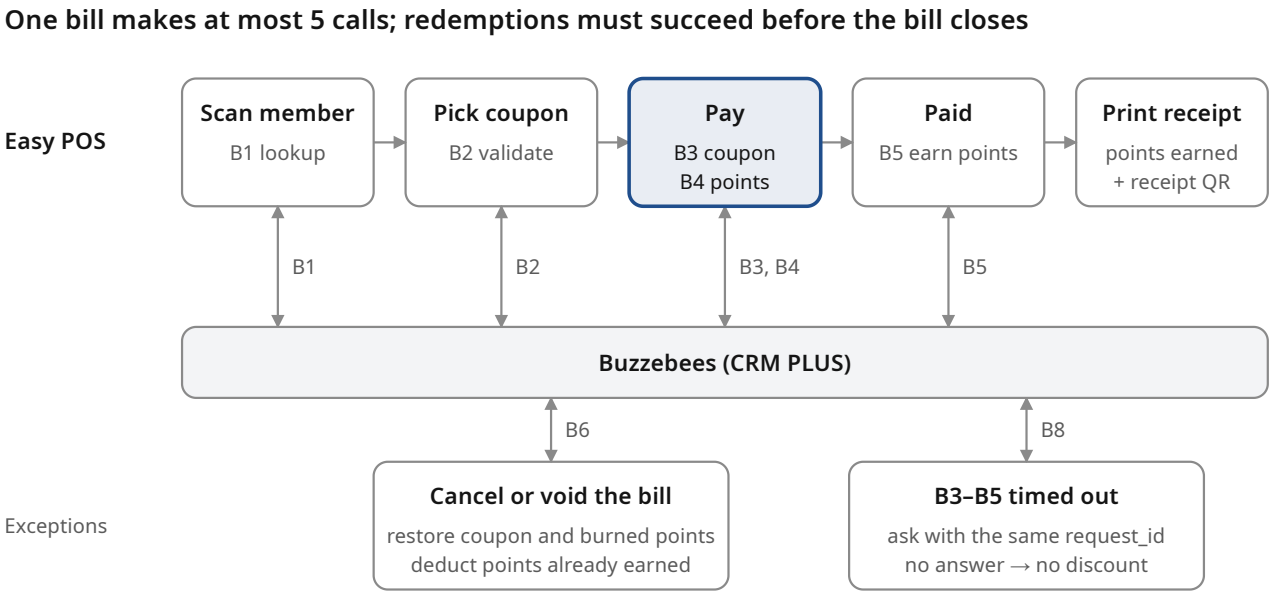
API	Name	Type	Caller → provider	Trigger	Timeout
B0	Health check	Support	Easy POS → Buzzebees	At shift opening, and every 5 minutes after a failed check	2 s
B1	Member lookup	Core	Easy POS → Buzzebees	Cashier scans the member QR or enters a phone number	3 s
B2	Validate coupon	Core	Easy POS → Buzzebees	Customer selects a coupon, before payment	3 s
B3	Redeem coupon	Core	Easy POS → Buzzebees	Payment is confirmed	3 s
B4	Burn points	Core	Easy POS → Buzzebees	Payment is confirmed	3 s
B5	Earn points	Core	Easy POS → Buzzebees	Payment succeeded, before the receipt prints	3 s (retryable)
B6	Void	Core	Easy POS → Buzzebees	Bill cancelled before closing, or voided / returned later	5 s (retryable)
B7	Master data	Support	Easy HQ → Buzzebees	Daily at 05:00, or when branches or menu items change	30 s
B8	Transaction status	Support	Easy POS → Buzzebees	B3–B5 timed out and the POS needs the outcome before closing	3 s
B9	Reconciliation report	Support	Oshinei reporting → Buzzebees	Daily at 06:00	30 s

Core APIs (B1–B6) sit directly in the payment path and therefore have short timeouts. Support APIs (B0, B7, B8, B9) serve operations, recovery and reconciliation.

5. Call sequence per bill

POS-side developer (call order and failure handling): Easy · Provider-side developer: Buzzebees

A bill makes at most five API calls (B1–B5), all initiated by Easy POS. Coupon redemption and point burning (B3, B4) must succeed before the bill closes. Point earning (B5) is called after payment succeeds and before the receipt prints, so the receipt can show points earned and the balance.



API call sequence per bill: five steps and two exceptions

1. The cashier scans the member QR from LINE or enters a phone number → **B1 member lookup**. The POS shows the name, point balance and available coupons.
2. The customer selects a coupon → **B2 validate coupon** (no deduction yet). The POS applies the discount returned.
3. The customer asks to burn points → the POS computes the discount from point_balance × point_value_thb returned by B1, with no API call.
4. Payment is confirmed → **B3 redeem coupon** and **B4 burn points**, one at a time. If either fails, the POS reverses the successful one with B6 and informs the cashier.
5. Payment succeeds → **B5 earn points** → the receipt prints with points earned, balance and the receipt QR. If B5 does not answer in time, the receipt prints "Points will be credited within 24 hours" and the POS places the request in the retry queue.
6. Bill cancelled before closing after B3/B4 succeeded → **B6** immediately with the transaction_refs to restore the coupon and points.
7. Void or return after closing → **B6** restores the coupon and burned points and deducts earned points. For a partial return, call B6 for the whole bill, then call B5 again with the post-return amount.

When B3, B4 or B5 times out: the POS calls **B8 transaction status** with the same request_id. completed → use that result; not_found → treat as not deducted and give no discount; B8 also unanswered → give no discount and close the bill at full price. Any deduction that may have happened appears in the reconciliation report, and Buzzebees restores it to the customer.

6. Request and response format

Final format owner: Buzzebees (aligned with the Easy HO_CRM specification)

All APIs use HTTPS with JSON payloads. Endpoints are prefixed with `/pos/v1/` (proposed). Buzzebees will provide separate sandbox and production base URLs.

HTTP headers on every request

Header	Value
X-Api-Key	The branch API key (Buzzebees issues one key per branch)
X-Branch-Code	Easy branch code; must match the API key
Api-Version	1
Content-Type	application/json; charset=utf-8

Common request body fields (B1–B7)

Field	Type	Description
request_id	string (UUID v4)	New for every request; the same value is reused when retrying after a timeout
branch_code	string	Easy branch code (BranchCode)
terminal_id	string	POS terminal
cashier_id	string	POS user who performed the transaction
request_time	string (ISO 8601)	POS time, e.g. 2026-10-14T18:32:05+07:00

Response envelope for every API

```
{
  "status": "ok",
  "code": "OK",
  "message": "สำเร็จ",
  "request_id": "6f1c0e5a-...",
  "server_time": "2026-10-14T18:32:06+07:00",
  "data": { }
}
```

Field	Description
status	ok or error
code	OK or an error code from the table in section 9
message	A short Thai message the POS can show to the cashier as is
request_id	Echo of the request value
data	The API-specific response fields; {} on error

HTTP status codes

Code	Meaning	POS action
200	Processed, both success and business rejection (see code)	Show the message
400	Malformed request	Log and report as a system defect
401	Invalid API key, or key does not match the branch	Show "Cannot connect to the membership system"
409	Same request_id with a different body	Generate a new request_id and resend
429	Rate limit exceeded	Wait for Retry-After, then retry
5xx or timeout	Buzzebees unavailable	Treat as failed and follow the no-response rules (section 9)

7. API details B0–B9

Provider-side developer: Buzzebees · Caller-side developer and Easy table writes: Easy (technical team)

Each API lists the fields the POS sends (in addition to the common fields), the fields Buzzebees returns in data, and the Easy table that stores them. Easy table names are taken from the current database schema and must be confirmed by Easy.

B0 Health check GET /pos/v1/health

No body; headers only. Returns status and server_time. On failure the POS shows "Membership system unavailable" and disables the coupon and burn buttons until the connection is restored.

B1 Member lookup POST /pos/v1/members/lookup

Field	Direction	Description
member_qr	Request	Code from the member QR in the LINE Official Account, valid for 5 minutes (send either member_qr or phone)
phone	Request	10-digit phone number, used when the customer cannot show the member QR
member_id	Response	Member ID, used in B2–B5
display_name	Response	Masked name, e.g. Som***, for the cashier to confirm with the customer
tier	Response	Membership tier (if any)
point_balance	Response	Point balance
point_value_thb	Response	Value of one point in baht; the POS uses it to compute the discount before calling B4
min_burn, max_burn_per_bill	Response	Minimum and maximum points that can be burned per bill
points_expiring_30d	Response	Points expiring within 30 days, for the cashier to mention
coupons[]	Response	Coupons valid at this branch: coupon_code, name, discount_text, expiry

Stored in Easy: Sale.CustomerId or Sale.CustRefNum = member_id (Easy to specify which). A phone lookup returns only the masked name and points, never other personal data.

B2 Validate coupon POST /pos/v1/coupons/validate

Checks whether a coupon applies to this bill without deducting it. May be called again whenever the bill changes.

Field	Direction	Description
coupon_code	Request	From the coupon QR, or chosen from coupons[] in B1
member_id	Request	Optional for general coupons not tied to a member
sale_type	Request	dine_in, take_away or delivery
bill_amount	Request	Amount before discount, including VAT
items[]	Request	item_code, qty, unit_price, amount, for item-level coupons
valid	Response	true or false
reason	Response	Error code when valid = false
discount_type	Response	amount (baht), percent or free_item
discount_value, max_discount	Response	Discount value and cap in baht
apply_to, item_codes[]	Response	bill or items, with the discounted item codes
easy_promotion_code	Response	The Easy discount code mapped to this coupon type; the POS records the discount with it

B3 Redeem coupon POST /pos/v1/coupons/redeem

Field	Direction	Description
coupon_code, member_id	Request	As in B2
receipt_no, sale_date	Request	With branch_code, forms the bill key
discount_amount	Request	Discount actually applied, in baht; Buzzebees verifies it does not exceed the B2 result
transaction_ref	Response	Transaction reference
redeemed_at	Response	Redemption time

Stored in Easy: PromotionByCRM (CouponNumber, DiscountAmount, TransactionRef, IsSuccess). A redeemed coupon becomes usable again only through B6.

B4 Burn points POST /pos/v1/points/burn

Field	Direction	Description
member_id, receipt_no, sale_date	Request	
points	Request	Points requested; must lie between min_burn and max_burn_per_bill
expected_discount	Request	Discount computed by the POS (points × point_value_thb); Buzzebees verifies it matches
discount_amount	Response	Discount granted
point_balance	Response	Balance after the burn
transaction_ref	Response	Transaction reference

Stored in Easy: Sale.RedeemedPoint, Sale.BalancePoint, plus one PromotionByCRM row for the point discount.

B5 Earn points POST /pos/v1/points/earn

Called after payment succeeds and before the receipt prints, so the receipt shows points earned and the balance. A bill key can earn only once (a second request returns DUPLICATE_RECEIPT).

Field	Direction	Description
member_id, receipt_no, sale_date, sale_time	Request	
sale_type, covers	Request	Sales channel and number of guests
gross_amount, discount_amount, service_charge, vat_amount	Request	Amount before discount, total discount, service charge and VAT, all separate so Buzzebees can compute points on the agreed base
net_amount_inc_vat, net_amount_ex_vat	Request	Net amount after discount, with and without VAT
payment_methods[]	Request	method, amount (cash, qr, credit_card, ...)
items[]	Request	item_code, item_name, qty, amount, for item-level bonus points
related_refs[]	Request	transaction_ref of B3/B4 on this bill, so Buzzebees can link them
earned_points	Response	Points earned on this bill
point_balance	Response	Point balance
transaction_ref	Response	Transaction reference
receipt_message	Response	Short message printed at the foot of the receipt (max. 2 lines)

Stored in Easy: Sale.EarnPoint, Sale.EarnPointNetAmount, Sale.BalancePoint; earned_points, point_balance and receipt_message are printed on the receipt. If B5 does not answer in time, the receipt prints "Points will be credited within 24 hours" and the POS queues a retry with the same request_id.

B6 Void POST /pos/v1/transactions/void

Used in three cases: cancelling a bill before closing after B3/B4 succeeded, voiding a closed bill, and returns. For a partial return, call B6 for the whole bill, then call B5 again with the post-return amount.

Field	Direction	Description
void_type	Request	cancel_before_close, void or return
receipt_no, sale_date	Request	The bill being voided (optional for cancel_before_close)
transaction_refs[]	Request	All B3–B5 transactions on the bill, including any from the receipt QR
void_reason	Request	Reason selected by the cashier in the POS
reversed[]	Response	transaction_ref and status (reversed, already_reversed, not_found)
point_balance	Response	Balance after the void; may be negative if the customer already spent points earned on this bill

Stored in Easy: PromotionByCRM.DiscountStatus of the reversed rows (Easy to confirm the column).

B7 Master data PUT /pos/v1/master/branches and PUT /pos/v1/master/items

Called by Easy HQ, not by branch terminals, sending a full snapshot each time. Buzzebees uses it to target coupons by branch and bonus points by menu item.

Field	Direction	Description
branches[]	Request	branch_code, name, active
items[]	Request	item_code, name, category, price, active
received	Response	Number of records accepted
rejected[]	Response	code and reason of each rejected record

B8 Transaction status GET /pos/v1/transactions/{request_id}

Used when B3, B4 or B5 times out and the POS must know whether Buzzebees already processed the transaction.

Field	Direction	Description
request_id	Request (path)	request_id of the request that timed out
status	Response	completed, not_found or reversed
original	Response	The original data of that response, when completed

POS action: completed → close the bill with the original result; not_found → treat as not deducted, no discount; B8 also unanswered → no discount, close at full price. Any deduction that may have happened appears in the reconciliation report and Buzzebees restores it to the customer.

B9 Reconciliation report GET /pos/v1/reports/daily?branch_code=&date=

Called by the Oshinei reporting system. Returns rows[] with the columns in section 10, plus total_count and generated_at. Uses a head-office API key, not a branch key. A CSV file by e-mail or SFTP is the fallback channel.

8. Receipt QR

QR format: Easy (technical team) · Decoding and point award: Buzzebees · Per-branch activation: Easy

Every receipt carries a QR code. Customers who did not identify as members at payment can scan it in the LINE Official Account to claim points later; new customers can sign up first and then claim. No POS API call is involved. (Easy already has the QrDataReceipt setting, currently disabled in all branches.)

QR content	Description
branch_code, receipt_no, sale_date, sale_time	Bill key; must match the values sent in B5
net_amount_inc_vat, earn_base_amount	Net amount, and the earn base agreed for point calculation
signature	HMAC-SHA256 digital signature over the fields above, using a secret key issued by Buzzebees (QRDataSecretKey in Easy), to prevent tampering with the amount (proposed)

The data layout inside the QR (separator, field order, encoding) shall follow what the Easy QRDataReceiptSetting supports. Easy will send the specification to Buzzebees.

Rules

- One receipt can claim points once, within 7 days of sale_date (proposed).
- For a bill already earned through B5, scanning the QR returns "Points have already been awarded for this bill" and awards nothing (checked by bill key).
- Receipt-QR transactions are recorded with channel = receipt_qr and carry a transaction_ref like any other.
- If a bill is voided after the customer scanned it, points are deducted through B6 (Buzzebees finds the transaction by bill key even when the POS holds no transaction_ref).
- A QR with an invalid signature is rejected and logged by Buzzebees in the reconciliation report.

9. General requirements

Compliance owners: Buzzebees (provider side and operations) · Easy (POS side: timeouts, retry queue, logging)

Security

- HTTPS with TLS 1.2 or higher on every API.
- One API key per branch, sent only in headers, never in the URL. A branch key cannot be used with another branch_code.
- Keys can be revoked per branch and rotated without downtime (old and new keys both valid for 7 days).
- Head-office keys for B7 and B9 should be restricted by source IP (IP allowlist) where possible.
- The only personal data passing through the API is member_id, the masked name, and the phone number in the B1 request.

Idempotency and duplicate prevention

- Same request_id + same body → the original result (HTTP 200), with no second execution.
- Same request_id + different body → HTTP 409.
- A bill key can earn (B5) only once, whether the request comes from the POS or the receipt QR.
- Retry queue for B5 and B6: retry after 1, 5 and 15 minutes, then hourly until 24 hours have passed; the item is then reported as outstanding in the reconciliation report.

Timeouts and Buzzebees unavailability

- The POS applies the timeouts in the API overview table; exceeding them counts as failure.
- The bill can always be closed; a membership outage must never block a sale.
- Coupons and point burning (B2–B4) are unavailable while offline; the cashier sees "Membership system unavailable" from B0.
- Points are credited later through the retry queue or when the customer scans the receipt QR; both use the same bill key, so nothing is duplicated.

Logging, time and currency

- Easy logs every request and response in CallAPILog and keeps them for 90 days; phone numbers in logs are masked as 08x-xxx-1234.
- All timestamps use ISO 8601 in Thailand time (+07:00).
- sale_date is the POS business date (SaleDate in Easy); bills after midnight but before the day is closed belong to the same business date.
- Amounts are in baht with 2 decimals, always split into VAT-inclusive and VAT-exclusive; points are integers.

Capacity requirements

Item	Value
Branches	Approximately 35
Bills per month (2026)	50,000–70,000, i.e. about 1,600–2,300 bills per day
Chain-wide peak hour	About 500 bills per hour (estimate), i.e. up to 2,500 requests per hour
Minimum throughput requested	10 requests per second
Availability	99.5% per month during trading hours 10:00–22:00 (proposed)

Environments and versioning

- A sandbox separate from production, with test members, test coupons and test API keys for Easy and Oshinei.
- The API version is carried in the Api-Version header. Breaking changes require at least 30 days' notice, and the previous version stays available for at least 6 months.
- Incident and planned-maintenance notification channels (see section 12).

Error codes (proposed)

Code	Meaning	Message shown by the POS
MEMBER_NOT_FOUND	Member not found	Member not found; suggest scanning the receipt QR to sign up
QR_EXPIRED	Member QR expired	Please ask the customer to refresh the member QR
COUPON_USED	Coupon already used	This coupon has already been used
COUPON_EXPIRED	Coupon expired	Coupon expired
COUPON_CONDITION	Conditions not met (branch, item, minimum spend, channel)	Coupon conditions not met, plus the Buzzebees message
COUPON_NOT_OWNER	Coupon belongs to another member	Coupon does not belong to this member
INSUFFICIENT_POINTS	Not enough points	Not enough points
BURN_LIMIT	Burn outside min_burn–max_burn_per_bill	X–Y points can be burned per bill
DUPLICATE_RECEIPT	This bill key has already earned points	Points have already been awarded for this bill
AMOUNT_MISMATCH	discount_amount or expected_discount differs from the Buzzebees calculation	Discount amount mismatch; please validate the coupon again
TXN_NOT_FOUND	transaction_ref to void not found	Transaction to void not found
AUTH_FAILED	Invalid key or key/branch mismatch	Cannot connect to the membership system; please inform the branch manager

10. Daily reconciliation

Reconciler: Oshinei (accounting) · Report provider and corrections: Buzzebees · POS-side data: Easy

Every day at 06:00 Oshinei pulls the previous day's transactions through B9 and compares them with Easy on three points: (1) every Buzzebees transaction has a matching bill in Easy; (2) every Easy bill with points or coupons has a Buzzebees transaction; (3) discount amounts and points agree on both sides.

Column	Description
transaction_ref, request_id	References on both sides
type	earn, burn, coupon or void
channel	pos or receipt_qr
branch_code, sale_date, receipt_no	Bill key
member_id	Member ID
points	Positive = earned, negative = burned or deducted
discount_amount, coupon_code	Discount in baht and coupon code (if any)
status	completed, reversed or rejected
created_at	Time recorded by Buzzebees

- Mismatches (present on one side only, or amounts differ) shall be resolved within 3 business days, with Buzzebees adjusting points to match the bill in Easy.
- Coupon and point discounts must be separable from other Easy discounts through PromotionByCRM, so the store's discount reporting stays clean.
- Items left in the retry queue beyond 24 hours are sent by Easy to Oshinei every morning for follow-up.

11. Acceptance test cases

Testing runs in the sandbox first, then again at one pilot branch with live bills. Every case must pass, and the reconciliation report for the test day must match Easy for every transaction.

Test group	Tester	Acceptance
Normal checkout, coupons and points, voids	Buzzebees with Easy (technical team)	Oshinei (project owner)
Network and retries	Easy (technical team)	Buzzebees confirms provider side; Oshinei accepts
Receipt QR, master data and reporting	Buzzebees	Oshinei (accounting reviews the report)
Pilot-branch retest	Easy (field team) with Buzzebees	Oshinei (branch operations)

Normal checkout

- [ ] B0: disconnect → the POS shows "Membership system unavailable" and recovers automatically when the connection returns.
- [ ] B1: lookup by QR and by phone returns the correct masked name, balance and branch coupons; expired QR → QR_EXPIRED.
- [ ] B5: a normal bill earns per the rules, the receipt prints points earned and balance, and Sale.EarnPoint matches Buzzebees.
- [ ] B5: a bill with discounts, service charge and split payment earns on the agreed base.
- [ ] B5: a delivery bill earns or does not earn, as agreed.

Coupons and points

- [ ] B2/B3: bill-level, item-level and free-item coupons apply and are recorded in PromotionByCRM with the correct easy_promotion_code.
- [ ] B2: used, expired, wrong-branch, below-minimum and other-member coupons → the error codes in the table.
- [ ] B3: the same coupon cannot be redeemed at another branch at the same time.
- [ ] B4: normal burn, burn above balance and burn outside min/max → the last two are rejected.
- [ ] B4: mismatched expected_discount → AMOUNT_MISMATCH with no deduction.

Voids

- [ ] B6: cancel before closing after B3/B4 succeeded → coupon usable again and points restored immediately.
- [ ] B6: void a bill with a coupon, a burn and an earn → all three are reversed.
- [ ] B6: void a bill whose earned points were already spent → balance goes negative as agreed.
- [ ] B6 + B5: partial return → net points match the post-return amount.
- [ ] B6: repeated void → already_reversed, no double reversal.

Network and retries

- [ ] Internet cut at bill closing → the bill closes, the receipt prints "Points will be credited within 24 hours", and points arrive once the connection returns, without duplicates.
- [ ] Resend the same request (same request_id and body) → same result, no duplicate.
- [ ] Same request_id with a different body → HTTP 409.
- [ ] B8: B3 times out after Buzzebees redeemed → B8 returns completed and the POS closes with that result; timeout before reaching Buzzebees → not_found.
- [ ] Branch A key used with branch B code → HTTP 401.

Receipt QR, master data and reporting

- [ ] Scan a normal receipt QR → points awarded; scan again → nothing; bill already earned through B5 → nothing; tampered QR → rejected.
- [ ] B7: full branch and menu snapshot → all received, and item-level coupons can reference item_code.
- [ ] B9: the test-day reconciliation report matches Easy for every transaction, including voids and receipt-QR claims.

12. Items requiring agreement

The following items must be settled before development starts, grouped by owner.

Before development Owners: Easy and Buzzebees

- [ ] **Easy:** deliver the HO_CRM specification and receipt QR format so Buzzebees can align field names and endpoints, and confirm the Easy columns used (Sale.CustomerId / CustRefNum, PromotionByCRM).
- [ ] **Easy:** confirm that HO_CRM supports all calls B0–B8, and whether any unsupported call requires additional development cost.
- [ ] **Buzzebees:** confirm support for receipt-QR scanning from a third-party POS, and the availability of a sandbox.

Business rules Decision owner: Oshinei (project owner)

- [ ] Earn base: amount after discount, with or without VAT and service charge, and the earn rate per baht.
- [ ] Value of one point when burned, minimum and maximum per bill, and point expiry.
- [ ] Whether delivery bills (Grab, LINE MAN) earn points.
- [ ] Whether coupons combine with other discounts on the same bill, how many per bill, and whether a coupon and a point burn can be used together.
- [ ] When a bill is voided after its earned points were spent: allow a negative balance, or deduct from the next earn.
- [ ] Who sets the receipt footer message (receipt_message) and how often it may change.

Configuration and operations Owners: Buzzebees and Easy; approval: Oshinei

- [ ] **Easy:** mapping each coupon type to an Easy discount code (easy_promotion_code), and who maintains that mapping.
- [ ] **Buzzebees:** issuing per-branch API keys (Buzzebees issues, Easy configures), and the rotation procedure for new branches or a leaked key.
- [ ] **Oshinei and Buzzebees:** the Data Processing Agreement (DPA), data residency and retention period.
- [ ] **Buzzebees:** development fee, monthly fee, Service Level Agreement (SLA) covering availability and resolution times, and the incident channel.
- [ ] **Oshinei:** pilot branches, sandbox start date and go-live date.